

## Trainings ▾

### General Information

The cylinder head group consists of the cylinder head, valves, valve guides, valve springs, valve seat inserts, crossheads, crosshead guides, and the injector sleeves. The exhaust valves are manufactured from a material that is capable of operating at a higher temperature than the intake valves. Do not install the intake valves in the exhaust valve location. The valve seat inserts for the exhaust and the intake valves are also manufactured from different materials. Do not use the intake valve seat in the exhaust location.

Mark, label, or tag the cylinder head parts such as crossheads, valves, and valve springs with the cylinder number and location at time of removal. This practice will prove to be a valuable aid in diagnosing any cylinder head or part failures and to make sure the reuseable parts are installed in their original locations.

The cylinder head casting used with a dry type exhaust manifold is different than the casting used with a wet type exhaust manifold. The casting used with the wet type manifold can be identified by the three slots located around the exhaust port. The cylinder head for the dry type exhaust manifold does not contain the slots.

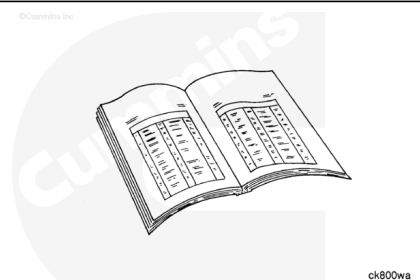
The fuel drain crossover was moved from the side of the cylinder head to the top of the cylinder head in March, 1987. Different styles can be interchanged by using fuel drain kits.

Oversize gaskets are available for the cylinder head and for wet and dry exhaust manifolds. These gaskets are used if the cylinder block deck height and/or the cylinder head thickness has been machined.

### Preparatory Steps

#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



#### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

**⚠ WARNING ⚠**

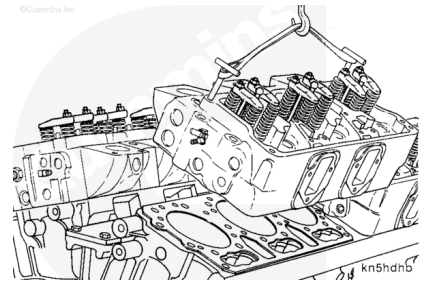
**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

- Disconnect the batteries. Refer to the OEM service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/15/15-008-018-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/15/15-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/15/15-011-007-tr.html>)
- Remove the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/15/15-010-002-tr.html>)
- Remove the valve push rods, injector push rod, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/15/15-004-014.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/15/15-003-011-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/15/15-003-013-tr.html>)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/15/15-006-026-tr.html>)

## Remove

### **⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



### **⚠ CAUTION ⚠**

**The cylinder head capscrews and washers are special parts. Do not damage the parts by throwing them against each other or in a container. Damage to the parts can result in engine damage.**

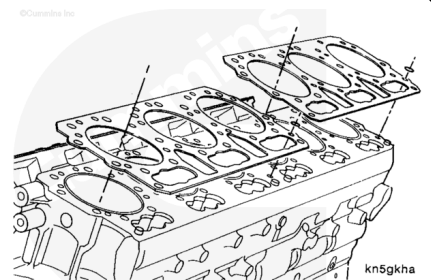
**Note :** There are 18 capscrews for each cylinder head.

Remove all the cylinder head capscrews and washers.

Remove all the cylinder heads.

### **⚠ WARNING ⚠**

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Remove all the capscrews.

Remove the plates and gaskets.

Discard the gaskets.

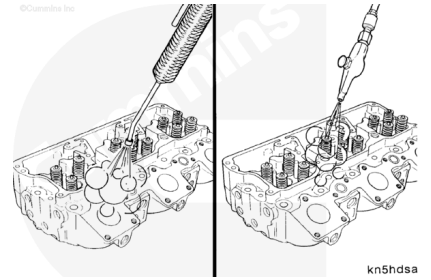
## Clean and Inspect for Reuse

### **⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

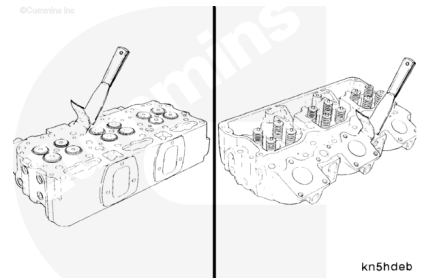


Steam clean the cylinder heads. Dry with compressed air.  
Blow out all capscrew holes, fuel passages, and oil passages.

Remove the gasket material from the rocker lever housing surface.

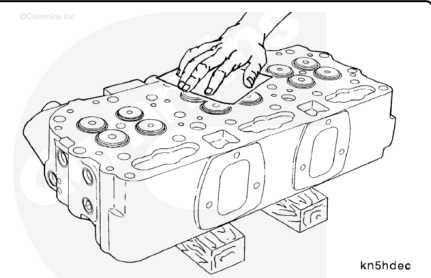


Use a gasket scraper to remove the heavy dirt and debris from the cylinder head gasket surface and the exhaust manifold gasket surface.



**⚠ WARNING ⚠**

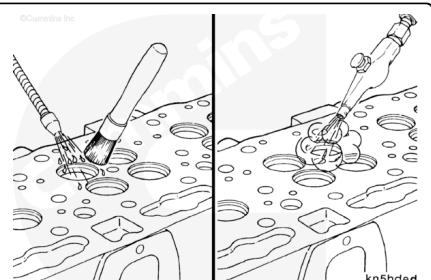
**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



Clean the combustion deck with a Scotch-Brite™ cleaning pad, or equivalent, and solvent.

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**



**⚠ WARNING ⚠**

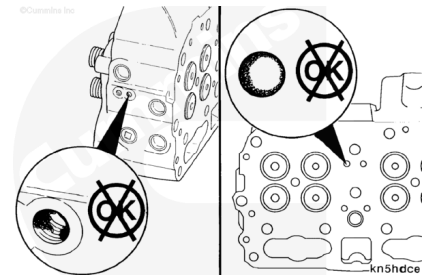
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean with solvent.

Dry with compressed air.

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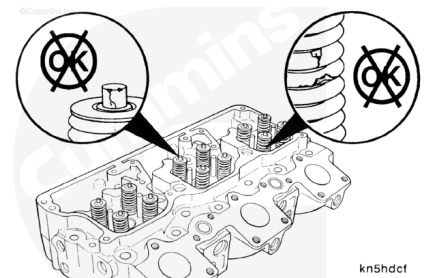


Check the fuel drillings and the water passages for restrictions or foreign material.

Remove any obstructions.

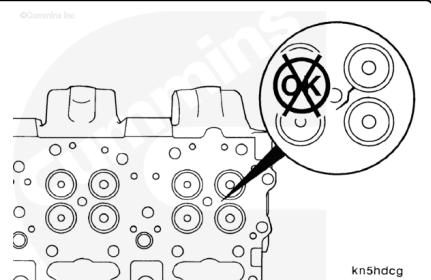
Check the valves and the valve springs for cracks, bent or broken valve stems, broken valve springs, or other damage.

**Note :** If cracked or damaged parts are identified, the cylinder head must be rebuilt.

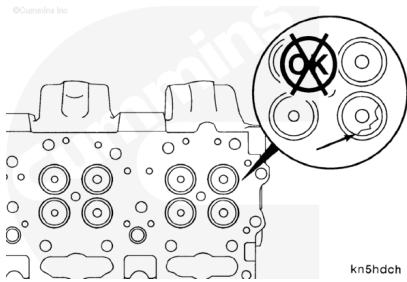


Check the cylinder head casting for cracks or other damage.

**Note :** If cracks or leaks in the cylinder head are suspected, complete a pressure test.

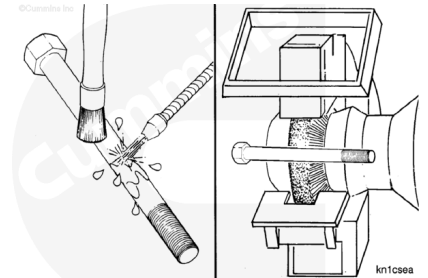


Check the valves for indications of leakage or burning. If indications of leakage or burning are found, the valves and the seats must be replaced or resurfaced.



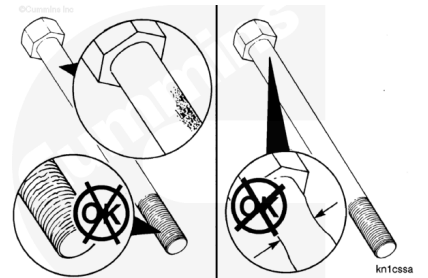
Use a petroleum-based solvent to clean the capscrews.

Clean the capscrews thoroughly with a wire brush, a wire wheel (soft), or use a non-abrasive bead blast to remove the deposits from the body and threads.



Inspect the cylinder head capscrews for damaged threads, corroded surfaces, or a reduced diameter (due to capscrew stretching).

**Note :** Do not reuse a capscrew that has damaged threads or a reduced diameter from having been stretched.



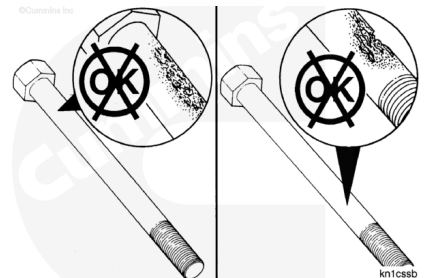
Do **not** reuse cylinder head capscrews with corrosion or pitting that exceeds 1 sq cm [0.155 sq in] in area.

Example:

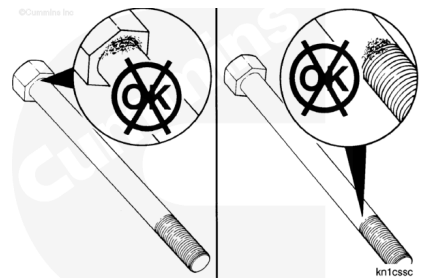
**Acceptable:** [3/8 x 3/8 in]

**Unacceptable:** [1/2 x 1/2 in]

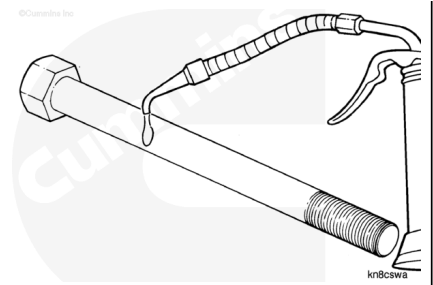
Do **not** reuse cylinder head capscrews with corrosion or pitting that exceeds 0.12 mm [0.005 in] in depth.



Do not reuse cylinder head capscrews if corrosion or pitting is located within 3.2 mm [0.125 in] of the fillet or threads.



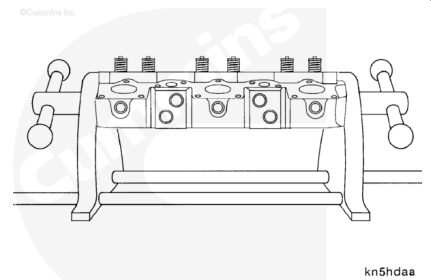
Immediately after cleaning and inspecting, apply a film of clean engine lubricating oil to capscrews that are to be used again.



## Disassemble

### **⚠ WARNING ⚠**

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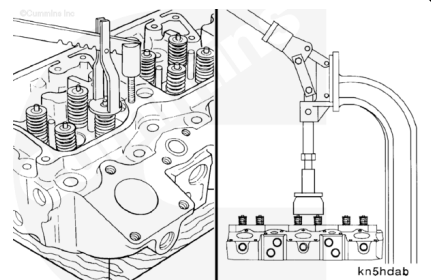


Install the cylinder head in the head holding fixture, Part Number ST-583, as illustrated.

**Note :** Do not install the cylinder head in the holding fixture if a valve spring compressor, Part Number ST-1022, is used.

### **⚠ CAUTION ⚠**

**The valve springs are under compression. Be careful when using the valve spring compressor. Personal injury can result if the tool slips from your hand.**



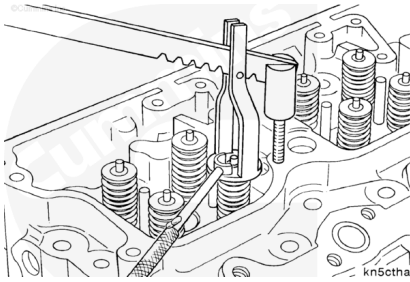
Use the valve spring compressor, Part Number ST-448, or an equivalent, to compress the valve springs.

**Note :** Install a wooden block between the valves and the head holding fixture to support valves.

**Note :** Valve spring stand, Part Number ST-1022, and the compressor plate, Part Number ST-1117, can be used to compress four springs at the spring time.

Use a pencil magnet to remove the valve collets. Discard the valve collets.

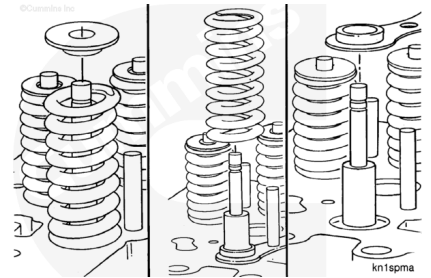
Slowly release the pressure on the valve spring.



Remove the valve spring retainer.

Remove the valve spring(s).

Remove the valve spring wear plate.



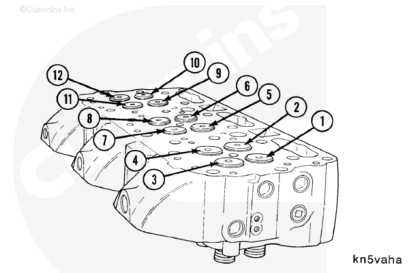
Remove the valves.

**Note :** Mark the location of the valves with an engraving tool during removal. The intake and exhaust valves are manufactured from different materials. The exhaust valves and seats can be installed in the intake valve location. Do not install the intake valves or seats in the exhaust location.

Repeat the procedure to remove the remaining valve collets, retainers, springs, wear plates, and valves.

Illustration call-outs are as follows:

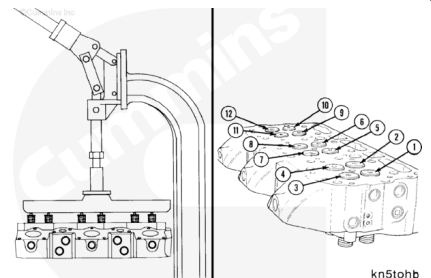
- Exhaust: 1, 3, 6, 8, 10, and 12
- Intake: 2, 4, 5, 7, 9, and 11.



## Pressure Test

### ⚠ WARNING ⚠

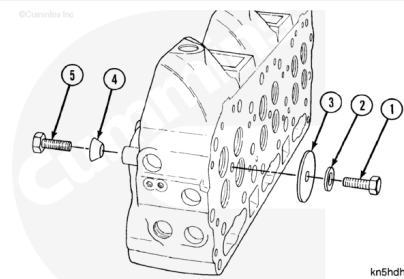
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the valves and the valve springs as follows:

1. Use valve spring compressor stand, Part Number ST-1022, and valve spring compressor plate, Part Number ST-1117.
2. Mark each valve as it is removed to identify its location in the cylinder head.

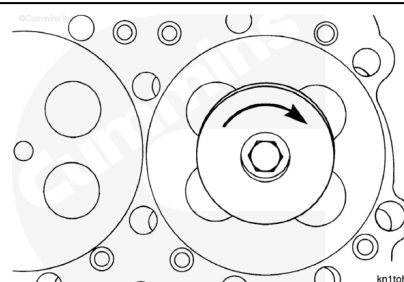
Install three injector sleeve holding tools, Part Number ST-1179, in each cylinder head.



Tighten the injector sleeve holding tool capscrews.

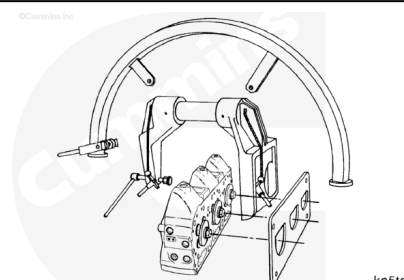
#### Torque Value:

1. 5 n•m [ 44 in-lb ]
2. 10 n•m [ 89 in-lb ]
3. 15 n•m [ 133 in-lb ]



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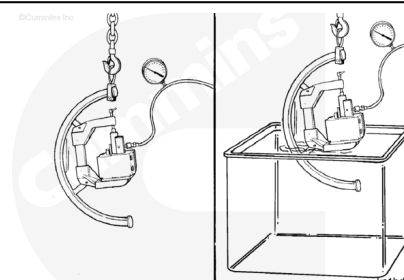
Install the cylinder head in a hydrostatic tester, Part Number ST-1012, and the hydrostatic tester adapter plate, Part Number ST-1021.

Connect a regulated air supply hose to the test fixture plate. Apply air pressure as follows:

#### Measurements

|              | kpa | psi |
|--------------|-----|-----|
| Air Pressure | 275 | 40  |

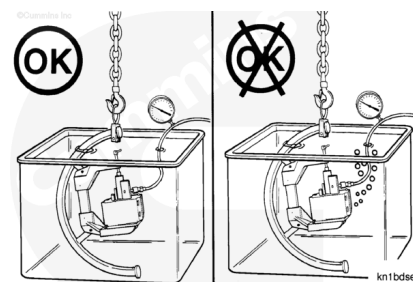
Use a hoist to put the cylinder head in a tank of warm water heated to 60°C [140°F].



Completely submerge the cylinder head in the water.

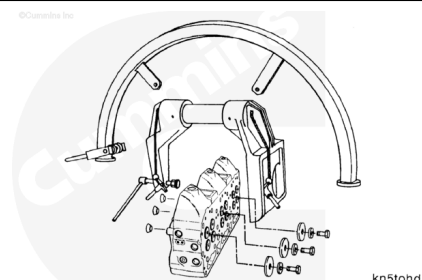
Inspect the air bubbles rising from the water.

If air bubbles are seen, replace or rebuild the cylinder head.



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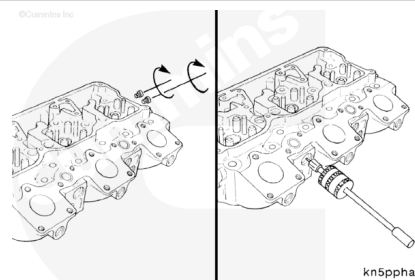
Remove the test equipment.

Dry the cylinder head with compressed air.

## Inspect for Reuse

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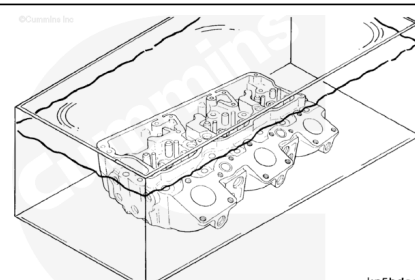
Remove the pipe plugs and the fuel fittings from the cylinder head.

Use a slide hammer to remove the expansion plugs.

**Note :** Expansion plugs must be removed from the head casting for cleaning purposes.

### **⚠ WARNING ⚠**

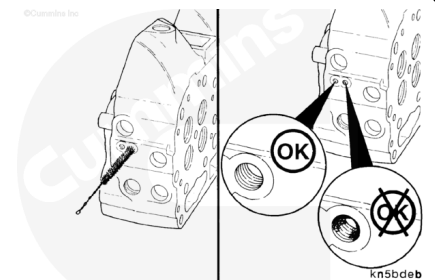
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**



Install the cylinder head and the parts in a tank of cleaning solution.

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**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

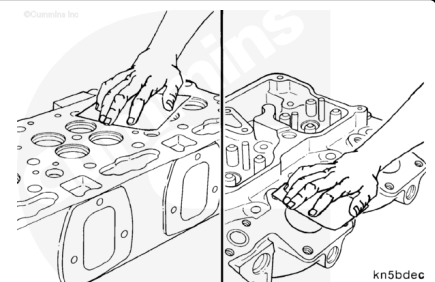
Remove the cylinder head and parts from the cleaning tank.

Use a fuel passage cleaning brush, Part Number ST-876, to clean the fuel and oil passages with solvent.

Dry with compressed air.

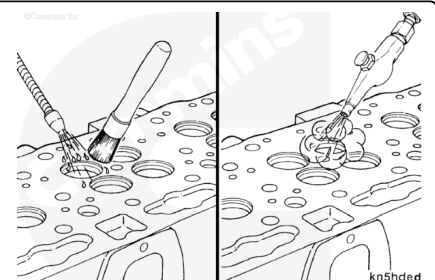
Inspect the fuel and oil passages to make sure they are clean.

Clean the combustion deck exhaust and intake manifold gasket surface with a Scotch-Brite®, or equivalent, and solvent.



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**⚠ WARNING ⚠**

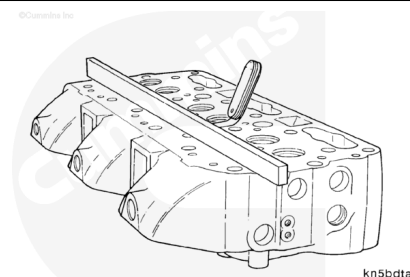
**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

Clean with solvent. Dry with compressed air.

Use a straight edge and a 0.10 mm [0.004 in] feeler gauge to measure the flatness of the cylinder head gasket surface.

| Flatness Per 25.4 mm [1.00 in] of Length |     |        |
|------------------------------------------|-----|--------|
| mm                                       |     | in     |
| 0.010                                    | MAX | 0.0004 |

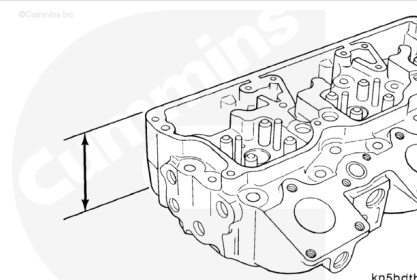
| Flatness Per Total Overall Length |     |       |
|-----------------------------------|-----|-------|
| mm                                |     | in    |
| 0.10                              | MAX | 0.004 |



**Note :** If the cylinder head is pitted, or has grooves or wear that are more than the maximum specified, the cylinder head must be resurfaced. Reference the Alternative Repair Manual, Bulletin 3379035.

Measure the thickness of the cylinder head.

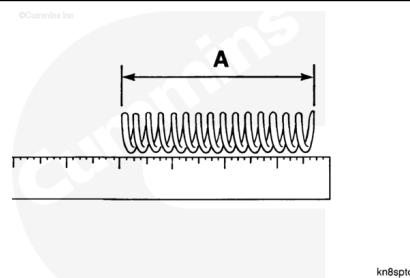
| Cylinder Head Thickness |     |       |
|-------------------------|-----|-------|
| mm                      |     | in    |
| 139.45                  | MIN | 5.490 |
| 139.95                  | MAX | 5.510 |



If the thickness is less than the minimum specification, do not use the head again.

Measure the free length of the valve spring.

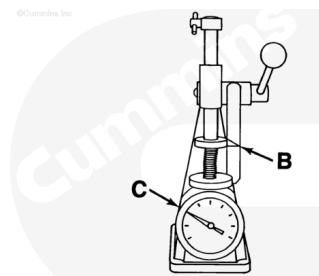
| Free Length (A)          |       |         |       |
|--------------------------|-------|---------|-------|
| Valve Spring Part Number | mm    | Limit   | in    |
| 211999                   | 68.20 | Nominal | 2.685 |



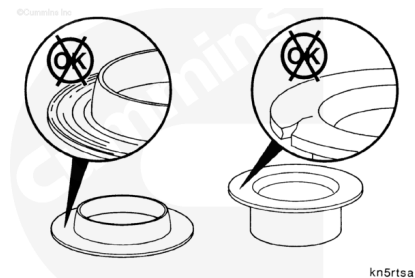
Use a valve spring tester, Part Number 3375182, to measure the valve spring load at the valve spring working height.

| Working Height (B)          |       |         |       |
|-----------------------------|-------|---------|-------|
| Valve Spring Part Number    | mm    | Limit   | in    |
| 211999                      | 43.81 | Nominal | 1.725 |
| Load for Working Height (C) |       |         |       |
| Valve Spring Part Number    | mm    | Limit   | in    |
| 211999                      | 631   | MIN     | 142   |
| 725                         |       | MAX     | 163   |

**Note :** If the valve spring load for the working height is less than the minimum specified, the valve spring(s) must be replaced.



Inspect the valve spring retainers and the valve spring guides for damaged or worn parts.

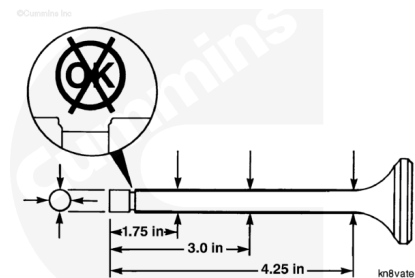


Inspect the valves for damage.

Inspect the collet grooves for wear.

Replace damaged valves or valves that have worn collet grooves.

Measure the outside diameter of the valve stem.

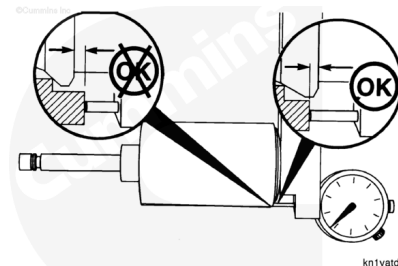


| Valve Stem Outside Diameter |     |       |
|-----------------------------|-----|-------|
| mm                          |     | in    |
| 11.40                       | MIN | 0.449 |
| 11.46                       | MAX | 0.451 |

**Note :** Valves that have a minimum stem diameter of 11.384 mm [0.4482 in] can be used if they are installed in new valve guides. Do not use valves with a minimum stem diameter of 11.384 mm [0.4482 in] in used valve guides. This can result in high blowby.

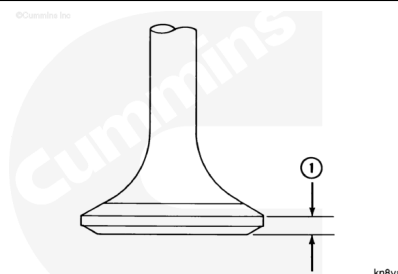
Use the valve head checking tool, Part Number 3823495, to measure the head thickness of the valves.

**Note :** Install the valve on the tool with the valve head contacting the tool as illustrated. If the valve head is even with, or above the end of the tool, the valve can be reground. If the valve head is below the end of the tool, the valve is not thick enough for regrounding.



If a valve head checking tool is **not** available, place the valve on a flat surface and measure the head thickness (1) at the outside diameter.

It is more accurate to use the valve head checking tool, Part Number 3823495.



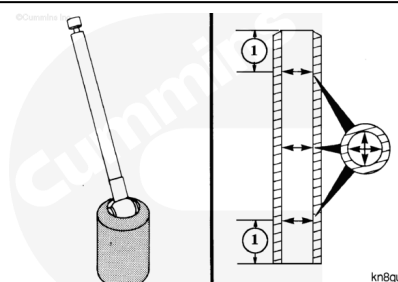
| Valve Head Thickness - Outside Diameter |     |       |
|-----------------------------------------|-----|-------|
| mm                                      |     | in    |
| 2.90                                    | MIN | 0.114 |

**Note :** If the valve head is worn thinner than the minimum specified, the valve must be replaced.

Inspect the valve guides for chips or cracks.

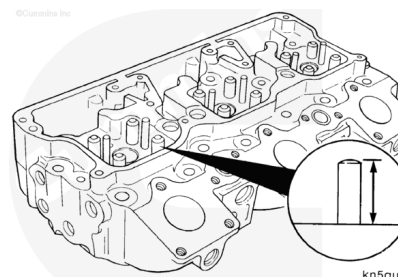
Measure the inside diameter of the valve guides 12.7 mm [0.50 in] from each end at the center as illustrated.

| Valve Guide Inside Diameter - Installed |     |        |
|-----------------------------------------|-----|--------|
| mm                                      |     | in     |
| 11.506                                  | MIN | 0.4530 |
| 11.557                                  | MAX | 0.4550 |



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Measure the valve guide installed height.

| Valve Guide Height - Installed |     |       |
|--------------------------------|-----|-------|
| mm                             |     | in    |
| 32.26                          | MIN | 1.270 |

### Valve Guide Height - Installed

| mm    |     | in    |
|-------|-----|-------|
| 32.77 | MAX | 1.290 |

**Note :** If damage is found or the valve guide(s) does not meet the limits specified, the valve guide(s) must be replaced.

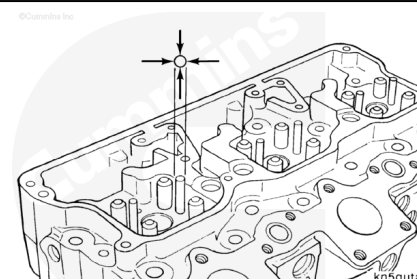
Inspect the crosshead guides for cracks or other damage.

Measure the outside diameter of the crosshead guides.

### Crosshead Guide Outside Diameter

| mm     |     | in     |
|--------|-----|--------|
| 10.975 | MIN | 0.4321 |
| 11.011 | MAX | 0.4335 |

**Note :** If damage is found or the crosshead guide(s) does not meet the limits specified, the crosshead guide(s) must be replaced.

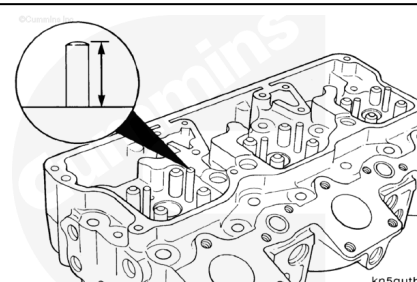


Measure the installed height of the crosshead guides.

### Crosshead Guide Height - Installed

| mm    |     | in    |
|-------|-----|-------|
| 47.25 | MIN | 1.860 |
| 47.75 | MAX | 1.880 |

**Note :** If damage is found or the crosshead guide(s) does not meet the limits specified, the crosshead guide(s) must be replaced.

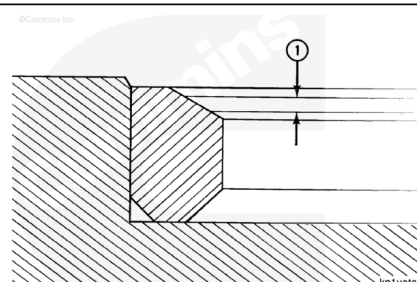


Measure the width of the valve seat area (1) of the valve seat inserts.

### Valve Seat Area - Width

| mm   |     | in    |
|------|-----|-------|
| 1.60 | MAX | 0.063 |
| 3.18 | MIN | 0.125 |

**Note :** If the valve seat area is worn wider than the maximum specified, replace the valve seat inserts. See



**⚠ CAUTION ⚠**

**Support the cylinder head to prevent damage to the injector tip that protrudes from the combustion face.**

Apply a bluing compound to the outside diameter of the injector at the injector seat area.

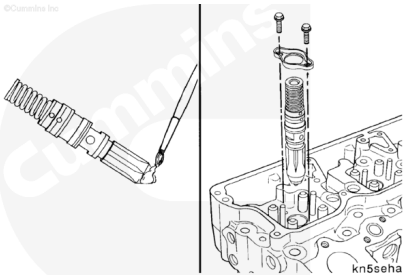
Install the injectors in the cylinder head without the o-rings.

Tighten the capscrews.

**Torque Value:**

[5/16 in] Capscrews

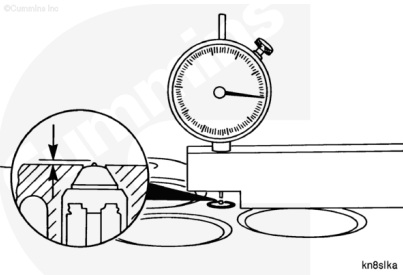
- 1. 6 n•m [ 53 in-lb ]
- 2. 12 n•m [ 106 in-lb ]
- 3. 18 n•m [ 159 in-lb ]



**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

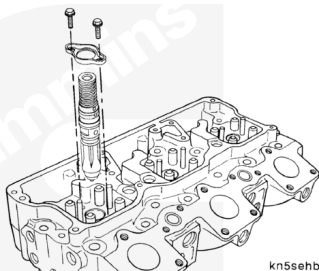
Turn the cylinder head over and use a gauge block, Part Number 3823495, and an indicator, Part Number ST-547-5, to measure the injector tip protrusion.



| Injector Tip Protrusion |     |       |
|-------------------------|-----|-------|
| mm                      |     | in    |
| 1.27                    | MIN | 0.050 |
| 1.52                    | MAX | 0.060 |

**Note :** If the injector tip protrusion does not meet the specifications given, the injector sleeve must be machined again or replaced.

Remove the injectors from the cylinder and check the pattern of the bluing compound in the injector sleeve.

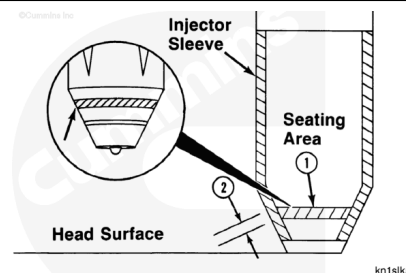


kn5sehb

The bluing pattern in the injector seating area (1) must be visible 360 degrees around the seating area.

The injector bore seating width (2) must be a minimum of 1.27 mm [0.050 in].

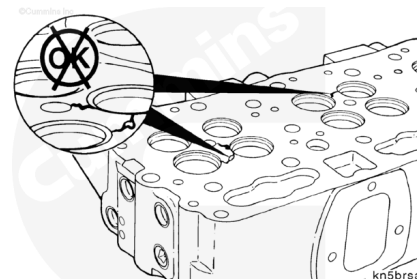
**Note :** If the injector protrusion is more than the maximum specified, or if the injector sleeve bluing pattern does not meet the specifications given, the injector sleeve (s) must be replaced. Refer to Procedure 002-014 in Section 2. (</qs3/pubsys2/xml/en/procedures/15/15-002-014.html>)



kn1slka

## Magnetic Crack Inspect

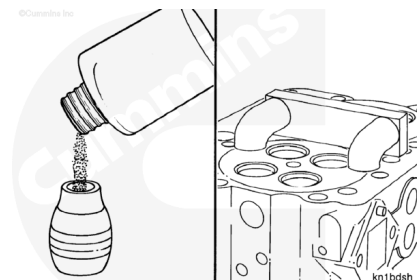
Use the magnetic crack detector, Part Number ST-1166, to find cracks in the areas around the valves and the injectors.



kn5braa

Fill the powder spray bulb  $\frac{1}{3}$  full of metal powder.

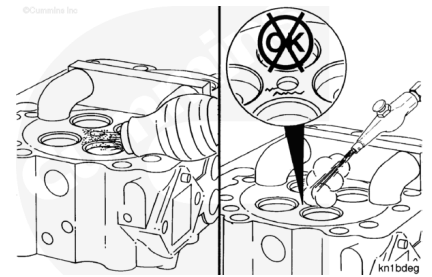
Place the magnetizing head on the cylinder head combustion surface to check for cracks that run across the head, as illustrated.



kn1bdsh

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**



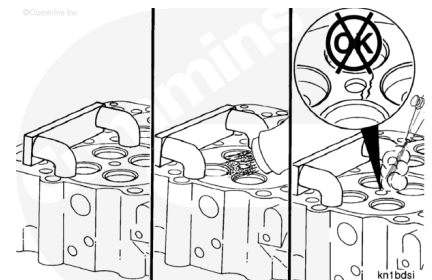
Lightly spray the metal powder onto the combustion surface.

Use compressed air regulated to 205 kPa [30 psi] to remove excess powder from the area. The powder will remain if there are any cracks and will appear as a white line.

Inspect the cylinder head for cracks in the combustion face.

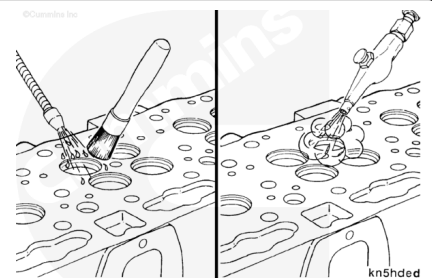
Place the magnetizing head on the combustion surface to check for cracks that run the length of the cylinder head, as illustrated.

Repeat the procedure as outlined above.



### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



### **⚠ WARNING ⚠**

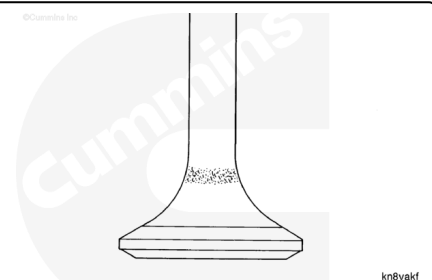
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Clean with solvent.

Dry with compressed air.

## **Exhaust Valves**

The exhaust valves contain two types of metal and must be inspected by the coil shot method. There will be a magnetic leakage apparent at the point where the two metals are welded together. The leakage will appear as a wide, fuzzy pattern of magnetic particles.

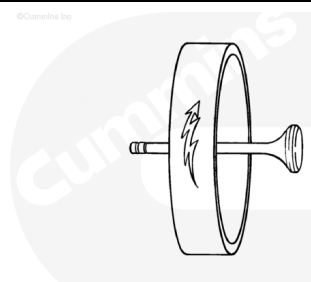


Use a magnetic particle method to check for damage.

Use the continuous method to apply coil shot amperage.

#### Amperage (Ampere Turns)

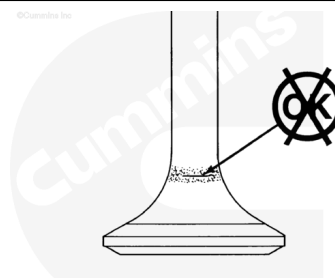
|     |      |
|-----|------|
| MIN | 3100 |
| MAX | 3500 |



kn8vakb

Use an ultraviolet light to check for indications of cracks.

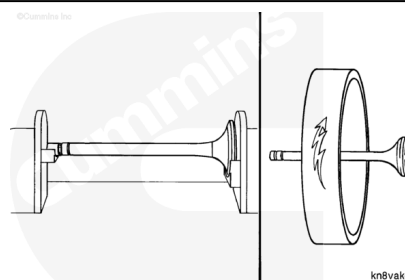
**Note :** A broad, fuzzy pattern will appear at the welded joint on the exhaust valves. If there is a distinct line in the pattern the, valve must be replaced.



kn8vakc

#### Intake Valves

**Note :** The intake valves contain only one type of metal and must be magnetized and inspected, using both the head and coil shot methods.



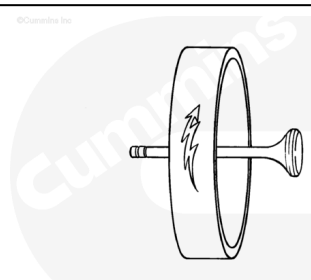
kn8vakg

Use the magnetic particle method to check for damage.

Use the continuous method to apply coil shot amperage.

#### Amperage (Ampere Turns)

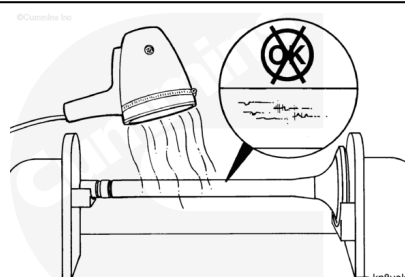
|     |      |
|-----|------|
| MIN | 3100 |
| MAX | 3500 |



kn8vakb

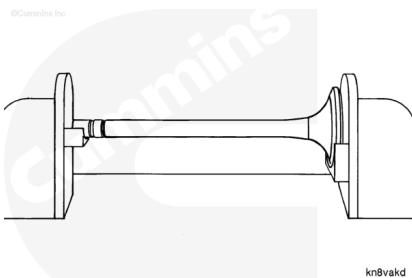
Use an ultraviolet light to check for indications of cracks.

The part must demagnetized between the head and coil shots.

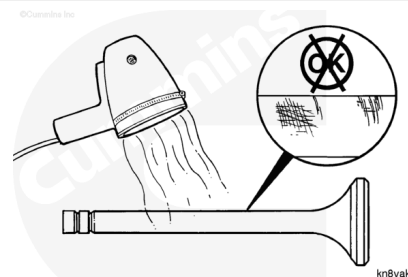


kn8vakh

Magnetize the intake valves again with a head shot at 500 to 700 amperes.



Use an ultraviolet light to check for indications of cracks.



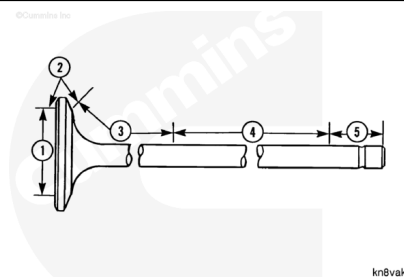
### Exhaust and Intake Valves Acceptance Criteria

In area (1), there **must** be **no** indication longer than 12.70 mm [0.500 in].

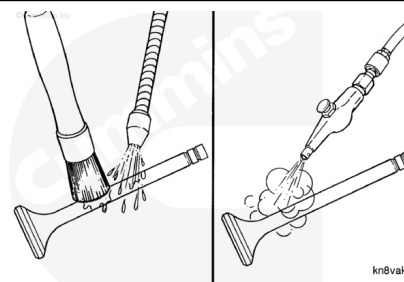
**Note :** There must be no more than five indications or no indications closer together than 3.18 mm [0.125 in].

Areas (2, 3, 4, and 5) must **not** have any magnetic indications or visible indications.

**Note :** A visible indication defined as a crack which can be seen through a three power (3x) magnifying glass after the magnetic particle suspension is removed.



Remove the magnetism and clean the acceptable valves.

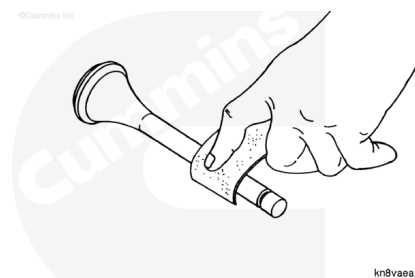


## Grind

Use Scotch-Brite™ to clean the valve stems.

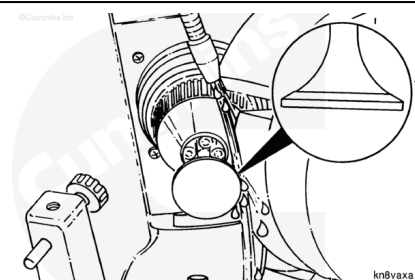
Clean the carbon deposits from the valve face and the head.

**Note :** The valves must be clean and free of carbon deposits before they are ground. Valves can be cleaned by the bead blasting method in the head area only. Bead blasting the stem will ruin the valves.



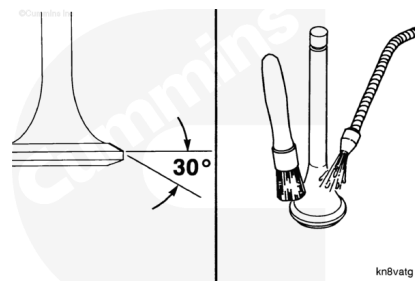
Use a valve facing machine, Part Number 3376256, to grind the face of the valve.

**Note :** Follow the instructions supplied with the valve facing machine for correct setup before grinding the valves.



**⚠ WARNING ⚠**

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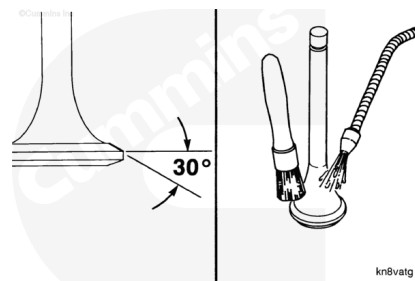
**⚠ WARNING ⚠**

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Grind the intake and exhaust valves to the angle illustrated.

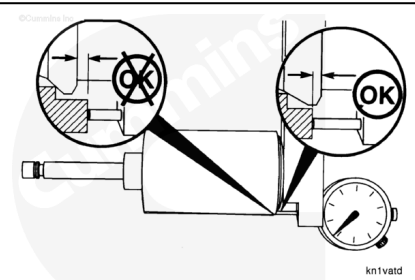
Use solvent to clean the metal particles from the valve.

Dry with compressed air.



Use a valve head checking tool, Part Number 3375933, to measure the head thickness of the valve.

**Note :** If the valve head is below the end of the tool, the valve must be replaced.

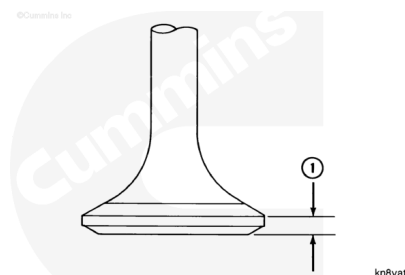


If a valve checking tool is not available, place the valve on a flat surface and measure the head thickness (1) at the outside diameter.

**Note :** This method is not as accurate as using the valve checking tool, Part Number 3375933.

| Valve Head Thickness (Outside Diameter) |     |       |
|-----------------------------------------|-----|-------|
| mm                                      |     | in    |
| 2.90                                    | MIN | 0.114 |

**Note :** If the valve head is worn thinner than the minimum specification, the valve(s) must be replaced.



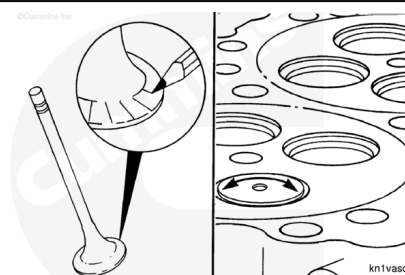
Use a lead pencil to mark across the valve face as illustrated.

Install the valve in the valve guide.

Hold the valve against the valve seat, and rotate the valve backward and forward three to four times. Correct contact against the valve seat will break the marks on the valve face.

**Note :** Valves and valve seats that are correctly machined do not require the use of lapping compound to make an air tight seal. If lapping compound is required, inspect the adjustments of the facing machine and the condition of the grinding stone.

All lapping compound **must** be cleaned from the parts before final assembly or quick wear-out will result.

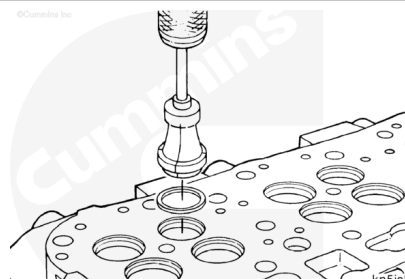


## Machine

### **⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

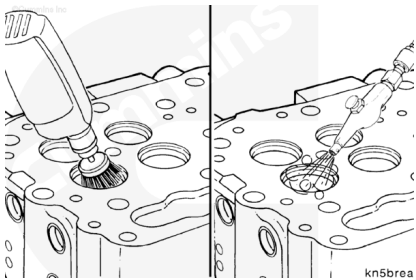
Use a valve seat grinding kit, Part Number 3376405, or equivalent, to machine the lower edge of the insert.



Use a slide hammer assembly, Part Number 3376617, and valve seat extractor, Part Number ST-1279-1, to remove the valve seat inserts from the cylinder head.

**⚠ WARNING ⚠**

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**⚠ WARNING ⚠**

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

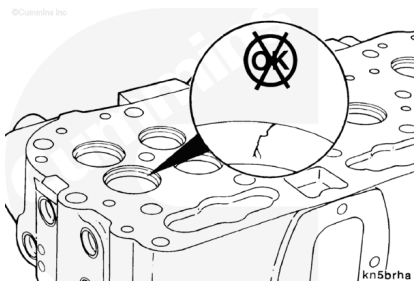
Use a wire brush and solvent to clean the deposits from the valve seat insert bores.

Use solvent to clean the cylinder head.

Dry with compressed air.

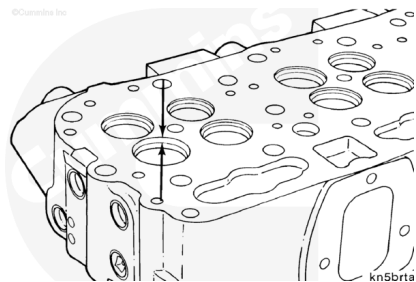
Inspect the insert bore for cracks or other damage.

**Note :** If cracks or other damage is found, it is possible to repair the cylinder head by machining the insert bore for oversize valve seat inserts. Refer to the Alternative Repair Manual, Bulletin 3379035.



Measure the valve seat insert bore depth in the cylinder head.

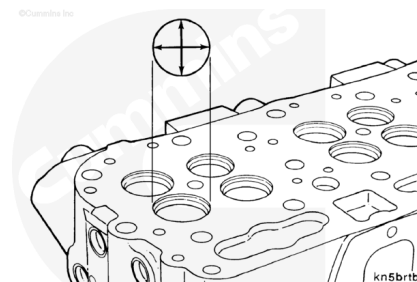
| Valve Insert Bore Depth (Standard Insert) |     |       |
|-------------------------------------------|-----|-------|
| mm                                        |     | in    |
| 7.24                                      | MIN | 0.285 |
| 7.37                                      | MAX | 0.290 |



**Note :** If the valve seat insert bore depth does not meet specifications given, it is possible to repair the cylinder head by machining the insert bore for oversize valve seat inserts. Refer to the Alternative Repair Manual, Bulletin 3379035.

Measure the inside diameter of the valve seat insert bore in the cylinder head.

| Insert Bore Inside Diameter (Standard Insert) |     |        |
|-----------------------------------------------|-----|--------|
| mm                                            |     | in     |
| 50.787                                        | MIN | 1.9995 |
| 50.813                                        | MAX | 2.0005 |



**Note :** If the valve seat insert bore inside diameter does not meet the specifications given, it is possible to repair the cylinder head by machining the insert bore for oversize valve seat insert. Refer to the Alternative Repair Manual, Bulletin 3379035.

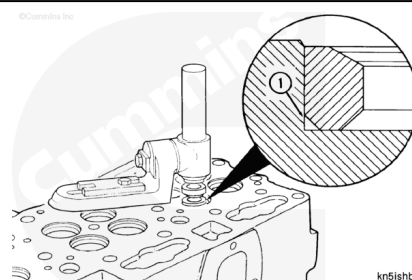
Install the base and swivel of the valve seat insert tool, Part Number ST-257, on the cylinder head to guide the valve seat driver.

**Note :** The insert chamfer (1) must be installed toward the bottom of the counterbore.

Make sure the intake and exhaust seats are installed in the correct location.

Use the valve seat driver and an arbor press to push the valve seat insert in to the counterbore.

**Note :** Make sure the insert is at the bottom of the counterbores.

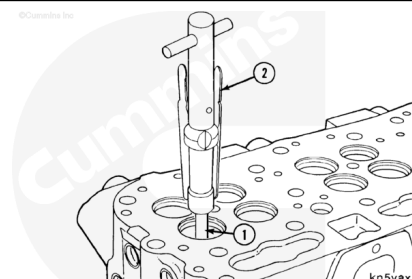


### **⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

**Note :** If the valve guide inside diameter exceeds the minimum worn limit, replace the valve guide before grinding the valve seat.

Use a valve seat grinding machine, Part Number ST-685, and valve guide arbor set, Part Number ST-663, when grinding the valve seat inserts. Install the valve guide arbor (1) in the valve guide with the arbor puller (2).



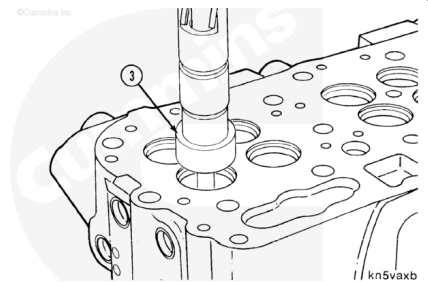
**Note :** Rotate the arbor to make sure it is correctly installed.

Install a valve seat grinding stone (3), Part Number ST-685-6, on the grinder unit.

**Note :** The grinding stone must be the correct size and have the correct angle ( [2 1/4 in diameter] x 30 degree grinding angle).

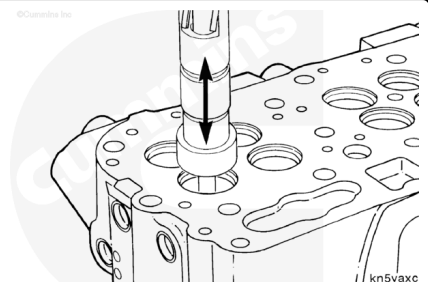
Install the grinder unit on the arbor.

**Note :** The grinding stone must not touch the valve seat insert when the drive unit motor is started.



Hold the drive unit in a vertical position. Use an up-and-down movement of 12.7 mm [0.50 in] travel and light pressure to grind the insert.

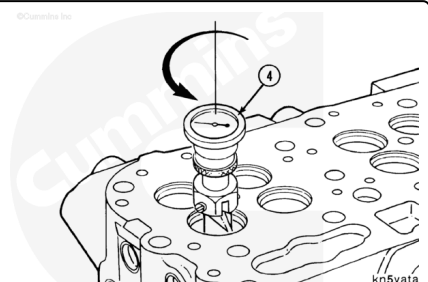
Remove the grinder unit from the arbor.



Install the eccentrimeter gauge (4), Part Number ST-685-4, on the arbor.

Measure the valve seat-to-valve guide concentricity.

| Valve Concentricity (Per 360 degrees) |     |       |
|---------------------------------------|-----|-------|
| mm                                    |     | in    |
| 0.05                                  | MAX | 0.002 |

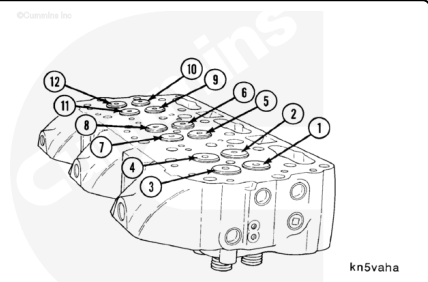


**Note :** If the valve seat concentricity does not meet the specifications, grind the valve seat again. If the specifications can not be met, replace the valve seat insert.

Install the reconditioned valves in their respective bores. Hold the valve firmly against the valve seat insert.

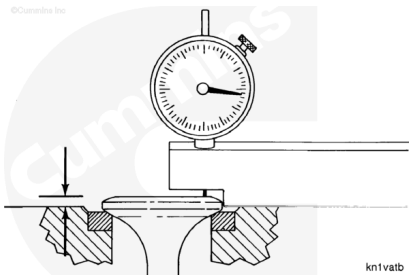
Illustration callouts are as follows:

- Exhaust: 1, 3, 6, 8, 10, and 12
- Intake: 2, 4, 5, 7, 9, and 11.



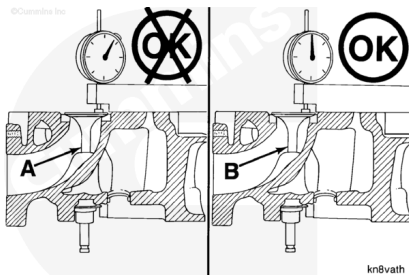
Use a gauge block, Part Number 3823495, to measure the valve protrusion.

| Valve Protrusion |     |       |
|------------------|-----|-------|
| mm               |     | in    |
| 2.92             | MAX | 0.115 |
| 3.91             | MIN | 0.154 |



If the valve protrusion is out of specifications, replace the old valve (a) with a new valve (B) before replacing the valve seat.

If the protrusion is within specifications with the new valve, proceed using the new valve with the old seat.

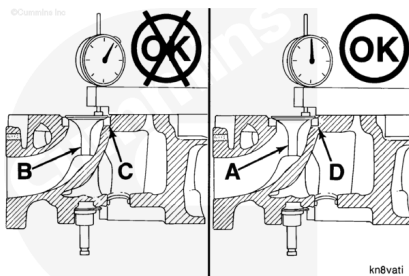


**⚠ WARNING ⚠**

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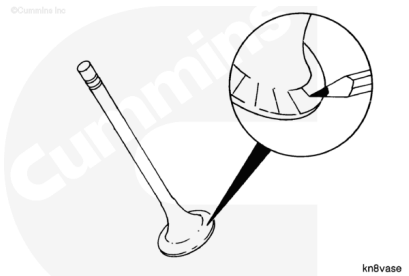
If the valve protrusion is still out of limits, even with the new valve (B), replace the seat (C) with a new valve seat (D). After replacing the seat, check the valve protrusion again with the old valve (A) to determine if it can be reused. If not, proceed using a new valve (B).

- (A) = Old valve
- (B) = New valve
- (C) = Old valve seat
- (D) = New valve seat.



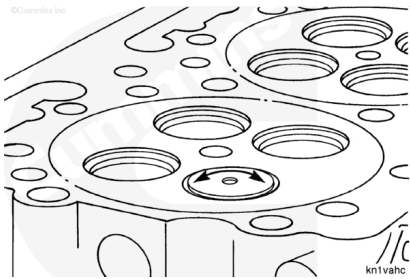
Check the valve seat contact area on the valve face to see if the valve seat contacts the center of the valve face.

Use a pencil or Dykem® to place vertical marks on the face of the valve.



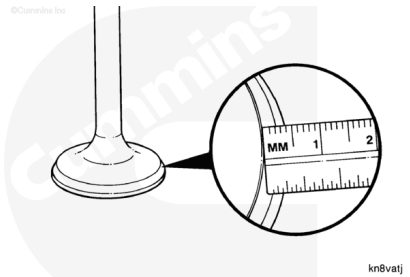
Install the valve in the valve guide. Hold the valve against the valve seat.

Rotate the valve backward and forward three to four times.

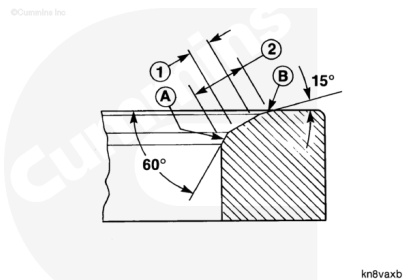


Remove the valve and measure the valve seat width and the seat contact area as indicated by the broken lines.

| Valve Seat Width Limit |     |       |
|------------------------|-----|-------|
| mm                     |     | in    |
| 1.60                   | MIN | 0.063 |
| 3.18                   | MAX | 0.125 |

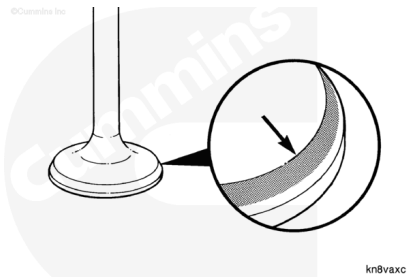


Grind area (A) with a 60 degree stone and area (B) with a 15 degree stone to center the seat on the valve face and to obtain the valve seat width limits.

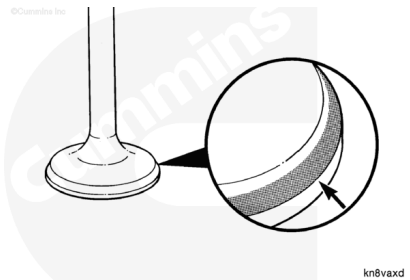


The location of the broken lines on the valve face is the determining factor of how much of each angle to grind.

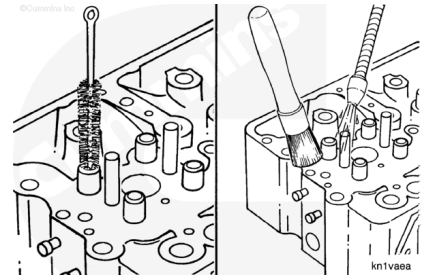
If the broken lines are at the bottom (next to the valve stem) of the valve face, the seat will require more grinding with the 60 degree stone than with the 15 degree stone.



If the broken lines are at the top of the valve face, the seat will require more grinding with the 15 degree stone than with the 60 degree stone.



**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



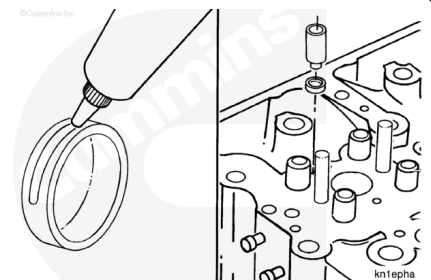
After grinding the valve seats, use a 12.7 mm [0.50 in] bristle brush. Clean the inside diameter of the valve guides.

Use solvent to clean the cylinder head. Dry with compressed air.

## Assemble

### **⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



Apply cup plug sealant, Part Number 3375068, to the outside diameter of the expansion plugs.

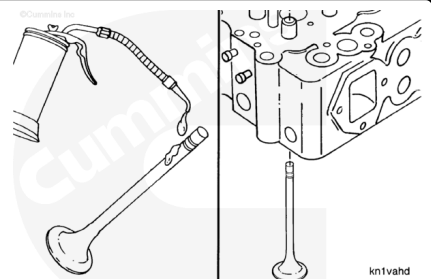
Use expansion plug drivers, Part Numbers 3376815, 3376816, and 3376817, to install the expansion plugs in the cylinder head.

**Note :** Use the correct expansion plug driver to make sure the expansion plugs are installed to the correct depth in the cylinder head.

Use clean 140W-EP oil to lubricate the valve stems. Do **not** use an excessive amount of oil for lubrication purposes.

Make sure the intake and exhaust valves are installed in the correct location.

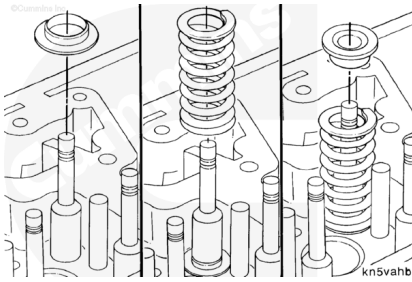
Install the valves.



Install the valve spring wear plates.

Install the valve springs.

Install the valve spring retainers.



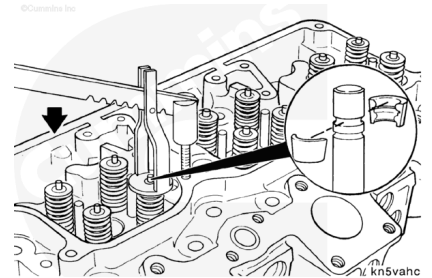
### ⚠ CAUTION ⚠

**The valve springs are under compression. Use caution when using the valve spring compressor as personal injury can result.**

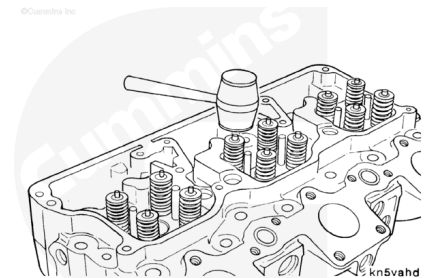
Use a valve spring compressor, Part Number ST-448, or equivalent, to compress the valve springs.

Install the new valve spring retainer collets.

**Note :** Use new valve spring retainer collets when rebuilding the cylinder head.



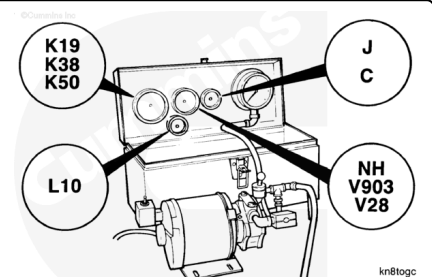
Use a mallet to gently hit the valve stems to make sure the valve collets are correctly seated in the spring retainer.



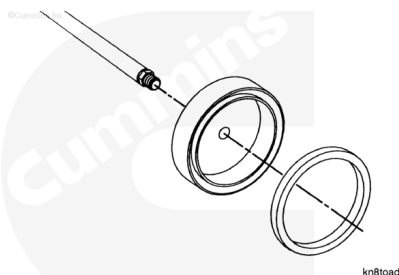
## Vacuum Test

Use a valve vacuum tester, Part Number 3824277, to inspect the seal between the valve and the valve seat.

**Note :** The valve vacuum tester can be used to test all Cummins® engine models. The seal ring, Part Number ST-1257-3, and the vacuum cup, Part Number ST-1257-6, are used on the V28 engine cylinder heads.



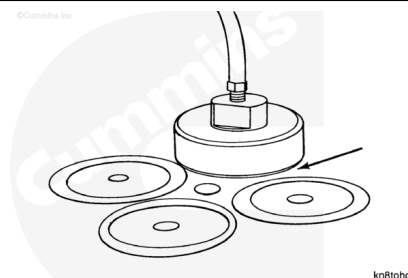
Install the seal ring and vacuum cup to the vacuum hose.



**Note :** The valves and valve seats must be clean and dry when vacuum testing.

Cover the valve with the cup and seal.

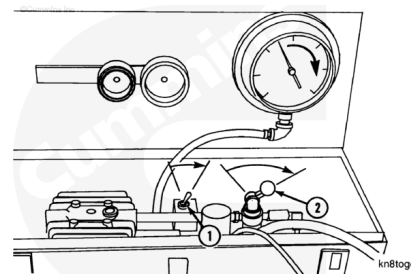
**Note :** The seal must make a tight contact on the cylinder head around the valve.



Move the toggle switch (1) to the ON position.

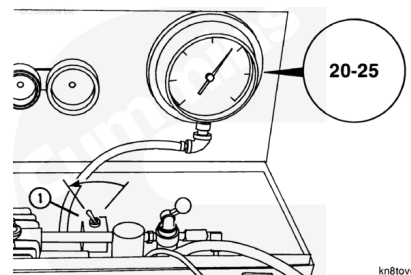
Turn the vacuum control valve (2) to the OPEN position.

**Note :** The seal must make a tight contact on the cylinder head around the valve.



Operate the vacuum pump until the gauge indicates the specified vacuum.

| Valve-to-Valve Seat Vacuum |     |    |
|----------------------------|-----|----|
| mm                         |     | in |
| 508                        | MIN | 20 |
| 635                        | MAX | 25 |

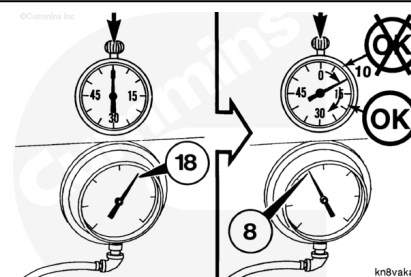


Turn the toggle switch (1) to the OFF position.

Use a stopwatch and begin timing when the needle on the gauge indicates 457 mm-Hg [18 in-Hg].

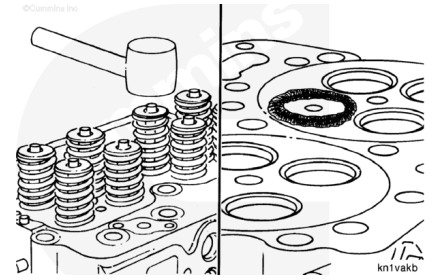
Stop timing when the needle on the gauge indicates 203 mm-Hg [8 in-Hg].

The elapsed time for the needle to move between the specified gauge readings **must** be 10 seconds or more.

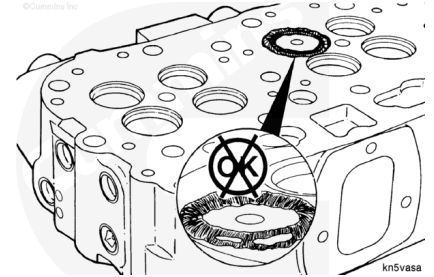


If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to make sure the equipment is functioning properly.
- Use a mallet to lightly hit the valve stem to make sure the valve is sealed. Repeat the test.
- Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.



**Note :** A break in the grease seal pattern will indicate leakage between the valves and the valve seat, or the valve seat insert and cylinder head.



## Install

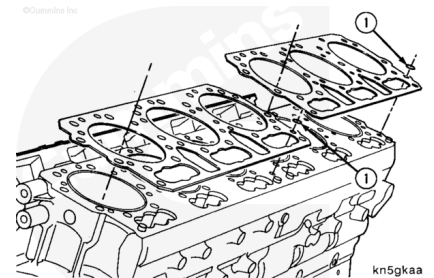
**Note :** The word TOP that is stamped on the head gasket must be visible after the gasket is installed.

**Note :** The gasket must slide easily over the dowel pins. If it does not, check the pins for damage.

Install the gaskets and one grommet on each cylinder head. Check to make sure the grommets remain in position.

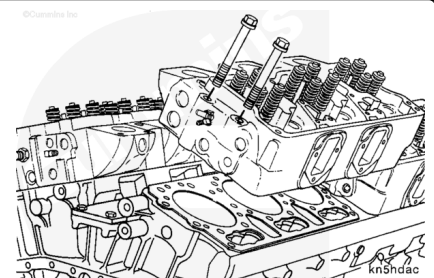
**Note :** An oversize head gasket must be installed on blocks that have had 0.50 mm to 0.76 mm [0.015 in to 0.030 in] removed from the top surface of the block. The plate and grommets are thicker on the oversize gasket.

**Note :** The water and push tube grommets are already in the gasket.



### **⚠ WARNING ⚠**

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Make sure the cylinder head fits over the dowels.

### **⚠ CAUTION ⚠**

**Allow the excess oil to drip from the capscrews before installation in the block or damage to the cylinder block can result.**

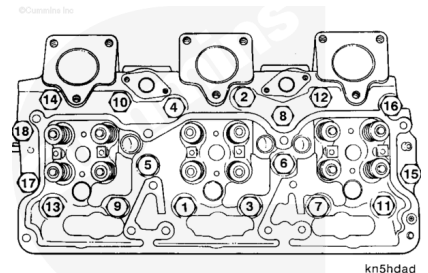
Use clean engine oil to lubricate the capscrew heads, threads, and washers.

Install the washers and capscrews.

Use the following steps to tighten the capscrews in the sequence illustrated.

#### **Torque Value:**

1. 45 n•m [ 33 ft-lb ]
2. 155 n•m [ 115 ft-lb ]
3. 290 n•m [ 214 ft-lb ]
4. 395 n•m [ 291 ft-lb ]



## **Finishing Steps**

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/15/15-006-026-tr.html>)
- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/15/15-003-013-tr.html>)
- Install the valve push rods, injector push rod, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/15/15-004-014.html>)

- Adjust the valves. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/15/15-003-004.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/15/15-003-011-tr.html>)
- Install the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/15/15-010-002-tr.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/15/15-011-007-tr.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/15/15-010-033-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/15/15-008-018-tr.html>)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.